Manufacturing and Production Processes

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- Manufacturing: From Latin "manu factus" (made by hand).
    - It is the process of converting raw materials into products.

Production Processes

Process	Simple Description	Key Features	Examples
1. Project- or Job-Based	One-of-a-kind production; only one unit is made at a time.	Highly skilled workers; frequent customer communication is essential.	Custom homes, haircuts, yachts.
2. Batch Production	Producing similar items in groups (batches); product moves stage by stage together.	Workers may only start and finish the process; machinery handles the main production.	Bakeries, textiles, furniture.
3. Mass Production	Creating standardized products in large quantities as economically as possible.	Heavy investment in machinery; workers mainly assemble component parts; generates high inventory.	Toilet paper, cell phones, automobiles.
4. Flow or Continuous Production	A process that runs twenty-four hours a day; products are homogeneous (identical).	Highly automated systems; workers act as monitors rather than active participants; reduces cost and increases efficiency.	Gas and oil, steel, chemicals.

Process Selection for Production

- Necessity: For any product, you must select the best manufacturing process from many options (e.g., casting, machining, molding).
- Decision Basis: The choice depends on many factors called process selection drivers.

Process Selection Drivers

- Quantity of the product.
- Cost (tooling, machines, maintenance, other costs).
- Time required for processing.
- Labor (skill level, supervision).
- Resources (energy consumption, material availability/cost).
- Product Requirements (dimensions/size, surface finish, design tolerances).
- Process Performance (capabilities, waste produced).

Steps for Process Selection

The selection is typically a four-step filtering and evaluation process:

STEP 1: Selection Criteria (Filtering)

- Establish criteria based on key drivers (volume, material, geometry, tolerances).
- Material is key: It immediately rules out many options (e.g., injection molding only works with polymers; die casting only with metals).
- Volume filters: High volume demands automated processes (manual labor is impractical).
- Geometry & Tolerances filter out processes that cannot achieve the required accuracy.

STEP 2: Identify Processes

- After Step 1 filtering, a smaller range of viable processes remains.
- Work with an experienced manufacturer to identify these processes that satisfy the material, quantity, and geometry needs.

STEP 3: Evaluate Processes

- Evaluate the remaining processes using less broad parameters:
  - Process capability (can it do the job?).
  - Processing time.
  - Tooling and equipment cost.
  - Degree of automation.
  - Skill required for operation.
  - Waste produced.
  - Post-processing required.
- Recommendation: Create a decision matrix to score or value each element.

STEP 4: Selection

- Use the weighted decision matrix (from Step 3) to identify the single best process.
- Give extra weight to the most important elements (e.g., cost or tolerance).
- Goal: Produce the part to the required standard at an acceptable cost.
- Note: Always work with experienced manufacturing engineers for best results.

Introduction to Casting

- Definition: A process based on the property of liquid to take the shape of the vessel containing it.
- Action: A cavity of the desired shape is made, contained within a mold.

History of Casting

- Ancient Process: Started 5000 years ago.
- Use: Making arrows, coins, knives, and very large structures (e.g., Jaivana, 50-ton cannon in Jaipur).

Basic Terms Used in Casting

1. Pattern: A replica of the desired final part.
2. Mold: The container with a cavity where the metal is poured. Divided into two halves: Cope (top) and Drag (bottom).
3. Gating System: A network of channels that directs molten metal from the pouring basin to the mold cavity.

Basic Steps in Casting

1. Pattern making
2. Mold making
3. Melting of metal and pouring
4. Cooling and solidification of metal
5. Cleaning of casting and inspection

Pattern

- Function: Replica of the desired product; used to form the mold cavity.
- Dimensions: Has slightly different dimensions than the actual part to compensate for allowances.

Pattern Materials

Material	Key Features	Main Use/Limitation
1. Wood	Easy to shape, low weight, low cost.	Most common (90%); absorbs moisture (distortion); low life (small quantity).
2. Metal	Longer life, closer dimensional tolerances.	Used for large quantity production. Aluminum is common.
3. Plastic	Low weight, smooth, durable, corrosion resistant.	Dimensionally stable (doesn't absorb moisture).

Material	Key Features	Main Use/Limitation
4. Polystyrene	Changes to gaseous state on heating.	Disposable Patterns (suitable for single, complicated castings).

Pattern Types

Type	Simple Description	Use Case
1. Single Piece	Pattern and cavity are fully contained in the lower flask (Drag).	Simple shaped and large castings.
2. Split Pattern	Models the part as two separate pieces along the parting line.	Parts are aligned by Dowel pins.
3. Loose Piece	Contains one or more loose parts attached by dowel pins/wire.	Used to make pattern removal easy; requires more skill (expensive).
4. Match-Plate	Both halves of the pattern are attached to opposite sides of a single plate.	Ensures proper alignment (of cope/drag/runner); used for larger production quantities.
5. Cope-Drag	Each half of the pattern is attached to a separate plate; molds are made independently.	Desirable for larger castings (assembled later).
6. Gated Pattern	Gate and runner patterns are permanently attached to the main pattern.	Used to manufacture multiple castings at once (mass production of small castings); saves time/labor.
7. Segmental	Pattern is a section moved to form a complete circular mold (e.g., wheel rim).	Used for circular castings.
8. Sweep	A sweep (board) is revolved around a spindle to form a large circular, symmetric mold.	Used for large circular molds.

Pattern Allowances

Allowances are intentional deviations from the final product dimensions provided on the pattern.

Allowance	Simple Purpose	Reason/Mechanism
1. Shrinkage/Contraction	Positive allowance (pattern size is increased).	Accounts for the metal's contraction/shrinkage after solidification (except grey cast iron).
2. Draft/Taper	Taper provided on vertical surfaces of the pattern ($\sim 1^{\circ}\text{ to }3^{\circ}$).	Allows the pattern to be easily removed from the sand mold without tearing the sides.

Allowance	Simple Purpose	Reason/Mechanism
3. Machining/Finishing	Extra material added to the pattern dimensions.	Provided only in machining areas to allow for finishing to the required size and surface quality.
4. Distortion/Camber	Pattern is designed with an intentional opposite distortion (camber).	Compensates for the uneven shrinking (internal stresses) that cause irregular castings ($\text{U}, \text{L}, \text{T}$ shapes) to distort.
5. Rapping/Shake	Pattern size is slightly reduced.	Accounts for the fact that the pattern is rapped (shaken) before withdrawal, which slightly enlarges the mold cavity.

Colour Coding for Pattern

- Purpose: Patterns are painted using shellac paints for protection and to identify their features.
- Scheme:

Colour	Indication (What the Colour Represents)
Black	Unfinished surface on the casting.
Red	Finished surface on the casting (needs machining).
Yellow	Core prints (where the core sets in the mold).
Black	Parting surface (the line where the mold halves meet).
Red/Yellow strips	Seats for loose pieces (areas where separate pattern pieces attach).

Types of Moulding Sand

Type of Sand	Simple Definition & Composition	Key Features & Use
i) Green Sand	Natural, moist sand (Silica sand, 18-30% clay, 6-8% water).	Fine, soft, porous, and damp. Used for small to medium castings. Avoids the cost/delay of drying.
ii) Dry Sand	Green sand that has been dried or baked to remove moisture.	Stronger and more compact structure. Used for large, heavy castings.
iii) Loam Sand	Thin, plastic paste of sand and up to 50% clay (milled with water).	Dries hard. Used for building moulds for large, regular-shaped castings (e.g., chemical pans). Dried slowly.
iv) Facing Sand	Used directly next to the pattern (contacts molten metal). Made of new silica sand and clay.	Must have high strength and refractoriness. Often mixed with carbon forms (plumbago) to prevent metal burn-in.
v) Backing Sand	Old, repeatedly used sand (black color). Used to fill the mould	Weak bonding strength due to repeated heat exposure. Also called Floor Sand or Black

Type of Sand	Simple Definition & Composition	Key Features & Use
	behind the facing layer.	Sand.
vi) System Sand	Used in machine moulding to fill the entire flask.	Must have higher strength, permeability, and refractoriness than backing sand.
vii) Parting Sand	Fine, sharp, dry, clay-free silica sand.	Used to prevent the two halves of the mould from sticking to each other or to the pattern.
viii) Core Sand	Used for making cores. Silica sand mixed with core oil (linseed oil, rosin, binders).	Provides strength for cores inside the cavity.
ix) CO_2-Sand	Silica grains coated with sodium silicate.	Packed around the pattern, then hardened instantly by passing CO_2 gas through it. Produces a strong, hard mould.
x) Shell Sands	Synthetic sand coated with resin (phenol/urea-formaldehyde).	Cured against a heated pattern to form a very strong, thin shell. No backing sand needed. Disintegrates after solidification.

Properties of Moulding Sand

1. Porosity (Permeability):
    - Ability of sand to allow gases to pass through.
    - Crucial for venting gases and steam generated when pouring molten metal.
    - Depends on particle size/shape, clay, binder, and moisture content.
2. Cohesiveness:
    - The strength of the sand; its ability to hold its particles together.
    - Important for retaining intricate mould shapes and preventing collapse during handling.
    - Improved by clay and bentonite.
3. Adhesiveness:
    - The property of sand to stick to the walls of the moulding box.
    - Enables the proper lifting of the cope (top half) along with the sand.
4. Plasticity:
    - The ability of the sand to flow into all corners and take a predetermined shape when rammed (compressed).
    - Ensures no spaces are left out around the pattern.
5. Flow-Ability:
    - The ability of the sand to free flow and fill the fine details and recesses of the pattern.
    - Varies with moisture content.
6. Collapsibility:
    - The property that allows the sand mould to collapse automatically after the casting has solidified.
    - Ensures the mold disintegrates with minimum force so the casting can be removed easily.
7. Refractoriness:
    - The ability of sand to withstand high temperature of molten metal without fusion/ softening
    - Sand moulds typically need to withstand temperatures up to 1650°C.

Meaning of Cores

- Definition: A pre-prepared shape of mold material (usually high-quality sand).
- Purpose: Used to create internal cavities, recesses, or projections in the final casting.

- Support: Core prints are added to the pattern to create impressions (seats) in the mold to support and hold the core at both ends.

Types of Cores

Type of Core	Simple Description	Key Feature / Use
i) Horizontal Core	Most common type; positioned horizontally at the parting surface of the mold.	Rests on seats provided by core prints at both ends; withstands turbulence of molten metal.
ii) Vertical Core	Placed vertically within the sand mold.	Top and bottom are tapered (top taper is greater).
iii) Balance Core	Extends only on one side of the mold.	Used when the casting has only a single side opening (e.g., producing blind holes or recesses); only one core print is used.
iv) Hanging Core	Suspended vertically in the mold.	No bottom support; held by hanging wires or a core collar resting in a cavity in the upper mold (Cope).
v) Drop Core	Used when the core must be placed above or below the parting line.	Also known as a wing core or chair core.
vi) Kiss Core	Used for multiple holes where less dimensional accuracy is required.	No core prints or seats are provided; the core is simply held approximately between the Cope and Drag.

Core Materials

- Composition: A mixture of sand, binders, and additives.
- Core Sands: Silica, zircon, olivine, etc.
- Core Binders: Core oils, resins, molasses, dextrin, etc.
- Constraint: Sand with more than 5% clay is unsuitable as it reduces both permeability and collapsibility.

Commonly Used Core Sand Mixtures

Core Type	Simple Composition	Key Features & Advantages
i) Core Sand	Green sand (small castings) or a mix of fire clay, green sand, and bentonite (heavier castings).	Cores are often oven baked to dry out moisture; dry sand cores are stronger. Rounded grains provide better permeability.
ii) Oil Sand	Sand mixed with core oils.	Provides good strength and excellent surface finish. Has better collapsibility after baking. Backed cores are hard for easy handling.
iii) Resin Sand	Sand mixed with thermosetting/thermoplastic binders (e.g., phenol, urea).	Offers very high strength, low gas formation, excellent collapsibility, and better dimensional accuracy.

Core Type	Simple Composition	Key Features & Advantages
iv) CO_2 – Sodium Silicate Sand	Silica sand mixed with sodium silicate ($\sim 3-4\%$).	Made hard instantly by passing CO_2 gas through it. Used for very large castings; no drying needed (fast method).

Core Components

Component	Function	Types
v) Core Binders	Holds sand grains together to provide binding strength.	Inorganic Binders (Fire clay, bentonite, iron oxide - fine powder). Organic Binders (Core oils, molasses, dextrin - harden rapidly).
vi) Core Additives	Used to improve special properties of the core.	For Stability: Kaolin/fire clay. For Hot Strength: Iron/aluminum oxide. For Refractoriness: Zircon/pitch flour. For Collapsibility: Raw dust (organic additives).

Core Chaplets

- Function: Used to support long cores when the end supports (core prints) are far apart.
- Purpose: Restricts the core from sagging (dropping) due to gravity during the pouring of hot molten metal.
- Material: Made of the same material as the casting metal so that they fuse and become an integral part of the final casting.

Types of Moulding Machines

Machine Type	Principle of Operation	Key Result / Use Case
1. Jolt Machine	The machine table lifts the flask a short distance and lets it fall freely under gravity (jolting).	Sand is packed evenly. Density is greatest around the pattern (best ramming where needed most). Used to create a strong mold.
2. Squeezing Machine	Compressed air or force is applied through a piston to squeeze the sand against a platen.	Ramming is best at the platen interface and not near the pattern (opposite of jolt). Limited to shallow work.
3. Jolt-Squeeze Machine	Employs a combination of the jolting and squeezing principles.	Provides the best results because it rams the sand near both the pattern and the platen; often used for pattern draw (roll-over).
4. Sand Slingers	Moulding sand is fed to a rapidly rotating impeller and thrown at high speed into the flask.	Can be used for flasks of any size (mass production or individual molds). Ensures high-density ramming across the flask.

Casting Defects

Defect	Simple Description	Key Causes	Simple Remedies
1. Blow Holes	Spherical, flattened, or elongated cavities inside the casting.	Rapid gas evolution from the mold, excess moisture, lack of porosity, hard ramming	Proper venting of molds/cores, avoid hard ramming, use proper moisture and grain size.

Defect	Simple Description	Key Causes	Simple Remedies
		of sand.	
2. Pin Holes	Small holes (less than 2 mm diameter) on or below the surface.	High pouring temperature, gas dissolved in molten metal, poor gas removal from metal, high moisture sand.	Maintain pouring temperature, increase flux proportion, reduce sand moisture, use effective venting.
3. Shift	External defect caused by misalignment of casting parts.	Misalignment of Cope and Drag, core displacement, worn clamping pins, pattern misalignment.	Repair/replace dowel/clamping pins, ensure proper alignment of pattern and mold halves.
4. Short Run (Misrun)	Mold cavity is incompletely filled (metal freezes before filling the whole cavity).	Insufficient metal supply, low fluidity of molten metal, low molten metal temperature, faulty gating system.	Adjust pouring temperature, adjust pouring rate, modify the gating system.
5. Hot Tears (Pulls/Hot Cracks)	Internal or external cracks that occur during solidification (often dark blue in appearance).	Lack of collapsibility of the core, high casting temperature, incorrect gating design, lack of fillets/corner radii.	Improved casting design (fillets), improved core collapsibility, proper solidification control, correct pouring temperature.
6. Cold Shuts	(Not detailed in the text, but included for completeness of list 6) Two streams of metal meet but do not fuse properly due to premature cooling.	Low metal temperature, slow pouring.	Increase pouring temperature and speed.

UNIT 2

Metal Forming Processes

- Manufacturing: The conversion of raw materials into finished products using various techniques (casting, forming, machining, joining, finishing).
- Metal Forming: A widely used manufacturing process that subjects material to plastic deformation to achieve a required shape and size.
- Action: The process applies forces like tension, compression, bending, or shear (or a combination).
- Products: Produces wrought forms (bulk or sheet) of materials.
- Industrial Importance (Advantages):
    - Cost effective.
    - Enhanced mechanical properties (e.g., increased strength).
    - Higher productivity.
    - Considerable material saving.
- Applications: Automotive (wheel rims, car bodies), aerospace, defense, and daily life items (cooking vessels, bicycle parts).

Classification of Metal Forming Processes

Metal forming processes are primarily divided into two main groups:

1. Bulk Forming (Bulk Deformation)
    - Raw Material: Raw materials have a low surface area to volume ratio (e.g., billets, bars).
    - Forces Applied: Primarily compressive force, or a combination of compressive, tensile, and shear forces.
    - Equipment: Accomplished using forming presses with a set of tool and die.
    - Processes:
        - Rolling
        - Forging
        - Extrusion
        - Drawing (wire or rod drawing)
    - Products: Gears, bushings, valves, connecting rods, engine parts.
2. Sheet Metal Forming
    - Raw Material: Works upon sheets, plates, and strips (high surface area to volume ratio).
    - Forces Applied: Predominantly tensile or shear forces.
    - Equipment: Mostly carried out in hydraulic or pneumatic presses using a set of tools called die and punch.
    - Processes:
        - Bending
        - Drawing (deep drawing)
        - Shearing
        - Blanking
        - Punching
3. Powder Forming (New Class)
    - Principle: Compacting and sintering metal powders.
    - Key Merit: Ability to produce parts very near to final dimensions (called near-net-shape forming) with minimum material wastage.
    - Material: Allows adjustment of material composition; sintered metals have greater formability than conventional wrought materials.
    - Challenge: The difficulty in completely eliminating porosity (empty spaces).

Forming Processes Classification

Forming is classified by the temperature at which deformation occurs relative to the material's recrystallization temperature (T_r).

1. Hot Forming (Hot Working)
    - Definition: Deformation carried out at temperatures above the recrystallization temperature ($0.5 T_m$ to $0.75 T_m$, where T_m is melting temperature).
    - Advantages:
        - Significant plastic deformation (large shape change) is possible.
        - Lower forces are required.
        - No strengthening due to work hardening (recrystallization cancels it out).
        - Materials with premature failure can be formed.
    - Disadvantages:
        - Shorter tool life (due to heat).
        - Poor surface finish.
        - Lower dimensional accuracy.
        - Sample surface oxidation.
2. Cold Forming (Cold Working)
    - Definition: Deformation generally done at room temperature or slightly above.

- Advantages (Over Hot Forming):
    - Closer tolerances (higher dimensional accuracy).
    - Good surface finish.
    - Higher strength and hardness due to strain hardening.
    - Desirable directional properties (grain flow).
    - Minimal heating costs.
    - Minimum machining required (near-net-shape forming).
- Disadvantages:
    - Higher forces and power are required.
    - Strain hardening limits the amount of forming possible.
    - May require repeated annealing-cold forming cycles.
    - Only possible if the workpiece is ductile enough.

Rolling

- Roll Forging: A process that reduces the cross-section of a cylindrical or rectangular rod by passing it through opposing grooved rolls.
    - Classification: Combines rolling and forging, but is classified as a forging operation.
    - Benefit: Produces stronger parts with desired grain structure compared to machining.

Forging

- Definition: The working of metal into a useful shape by hammering or pressing (oldest metalworking art).
- Action: Involves plastic deformation of material between two dies.
- Scale: Capable of making parts from small bolts to large turbine rotors.
- Temperature: Most operations are carried out hot, though some metals are cold forged.

Forging Types

- Open Die Forging (Smith Forging):
    - Metal is compressed by repeated blows (mechanical hammer) between flat-faced dies.
    - Shape is manipulated manually (depends on smith's skill).
    - Accuracy is lower than closed die forging.
    - Tooling is simple and inexpensive (allows large variety of shapes).
- Closed Die Forging (Impression Die Forging / Drop Forging):
    - The desired configuration is obtained by squeezing the workpiece between two shaped and closed dies.
    - Drop Forging: The ram (upper die) falls and strikes the workpiece, forcing the metal to fill the die contours.
    - Press Forging: Uses a relatively slow squeezing action instead of heavy blows, which penetrates deeply and gives the metal time to flow.

Forging Stages (Example: Connecting Rod)

1. Bar stock
2. Fullering (thinning the stock in certain areas)
3. Edging (distributing metal laterally)
4. Rolling
5. Blocking (initial shaping in the die)
6. Finishing (final shape impression)
7. Trimming (removing excess metal/flash)

Typical Forging Defects

- Pitted surface: Due to oxide scales sticking to the dies.
- Surface cracking: Due to temperature difference (surface vs. center) or excessive working at too low a temperature.
- Cold shut or fold: Flash/fin from a prior step is forced into the workpiece.

- Flash line crack: Occurs after trimming; remedied by increasing flash thickness.
- Internal cracking: Due to secondary tensile stress.

Extrusion Process

- Definition: A simple process where molten metal/polymer is forced through a shaped die using pressure.
- Mechanism: Pressure is generated by a rotating screw against the barrel wall.
- Materials Used: Most types of metals, thermoplastics, and rubbers.

Components of an Extruder

1. Drive
2. Gearbox and Thrust bearing
3. Feed hopper
4. Barrel

Advantages, Disadvantages, and Limitations

Category	Simple Feature	Explanation
Affordability	Low Cost & High Profit	Low costs compared to other molding processes; used for bulk production.
Material Reuse	Reduced Waste	Uses thermoplastics (can be repeatedly melted/hardened), allowing leftover material to be reused.
Flexibility	Consistent Cross-Section	Excellent for manufacturing products with a consistent cross-section (e.g., decorative trims).
Flexibility	Product Variation	Minor alteration allows for plastic sheets or products mixing plastic attributes (hard/soft surfaces).
Disadvantage	Size Variance (Die Swell)	Hot plastic expands after leaving the extruder (die swell). The exact expansion is hard to predict, leading to accepted deviations in product dimensions.
Limitation	Product Shape Limits	Cannot easily produce products that change cross-section (e.g., soda bottles that narrow for a cap).

Metal Spinning (Spin Forming)

- Definition: A metalworking process where a metal disc or tube is rotated at high speed and formed into an axially symmetric part.
- Principle: It is a forming (molding) process, not a material removal process (like machining). The sheet material is gradually formed over an existing shape (block).
- Method: Can be performed by hand or by a CNC lathe.
- Materials: Virtually any ductile metal (e.g., aluminum, stainless steel, high-temperature alloys).
- Size Limitation: Only limited by the size of the available equipment.

The Spinning Process

1. A formed block is mounted on a lathe's drive section.
2. A pre-sized metal disk (workpiece) is clamped against the block by a pressure pad.
3. The block and workpiece are rotated together at high speeds.
4. A localized force is applied to the workpiece (using levered tools) to cause the metal to flow over the block.
5. Shape Change: The metal must thicken, elongate radially, or buckle circumferentially since the final diameter is always less than the starting diameter.

6. Removal: Simple parts are removed directly; complex shapes may require a multi-piece block or an ice form (which melts away).

Applications

- Artisan: Specialty lighting, decorative household goods, architectural details.
- Commercial: Rocket nose cones, cookware, gas cylinders, brass instrument bells, public waste receptacles.

Hydroforming Process

- Definition: A specialized type of die forming that uses high-pressure hydraulic fluid to press room-temperature working material (ductile metal) into a die.
- Purpose: Shapes metals into lightweight, structurally stiff, and strong pieces.
- Materials: Virtually all metals capable of cold forming, including aluminum, brass, and stainless steel.
- Key Application: Automotive industry (e.g., creating strong, light, and rigid vehicle frame rails).

Working Mechanism (Tube Hydroforming)

1. A hollow tube is placed inside a negative mold (die) that has the shape of the desired part.
2. High-pressure hydraulic fluid is injected inside the tube.
3. The pressure forces the tube to expand until it conforms exactly to the complex shape of the mold.
4. This allows for complex shapes with concavities that are difficult for traditional stamping.

Advantages

- Design Freedom: Allows for the creation of highly complex parts.
- Part Reduction: Reduces the number of individual parts needed, minimizing assembly and welding.
- Weight Reduction: Creates a higher stiffness-to-weight ratio (stronger and lighter).
- Accuracy: Provides high dimensional accuracy and reduced springback.
- Durability: Results in higher mechanical strength and longer durability.
- Surface: Forming with a rubber element avoids the scratches associated with traditional stamping.

Disadvantages

- Slow cycle time.
- Expensive equipment.
- Lack of extensive knowledge base for tool and process design.

Introduction to Die Casting

- Definition: A variation of metalcasting where liquid metal is injected into a reusable steel mold (die) very quickly with high pressures.
- Distinguishing Feature: Uses reusable steel tooling and high-pressure injection.

Types of Die Casting

1. Hot-Chamber Die Casting (Gooseneck Casting)

- Process: The injection mechanism's cylinder chamber is fully immersed in the molten metal bath (melting pot is in the machine). A gooseneck system feeds the metal.
- Focus: Production efficiency (quick and convenient injection).
- Limitation: Direct immersion causes increased corrosion.
- Best Suited For: Metals with low melting points and high fluidity.
- Examples: Lead, Zinc, Magnesium, Copper.

2. Cold-Chamber Die Casting

- Process: The melting pot is outside the machine. The molten metal is ladled (automatically or by hand) into the injection system.
- Focus: Minimizing machine corrosion (eliminates immersion).
- Best Suited For: Applications that are too corrosive for hot-chamber, such as metals with high melting temperatures.
- Examples: Aluminum and Aluminum alloys.

Advantages

- Accuracy: Excellent dimensional accuracy (e.g., 0.1 mm for the first 2.5 cm).
- Finish: Smooth cast surfaces and thinner walls (approx. 0.75 mm) than sand casting.
- Complexity: Allows inserts (e.g., threaded inserts, heating elements) to be cast-in.
- Production: Rapid production rates.
- Finishing: Reduces or eliminates secondary machining operations.
- Strength: High casting tensile strength (up to 415 MPa).

Disadvantages

- Weight/Size Limits: Casting must be between 30 grams and 10 kg, and smaller than 600 mm.
- Cost: High initial cost (expensive equipment and tooling).
- Porosity: A certain amount of internal porosity (small voids) is common.
- Design Limits: Thickest section should be less than 13 mm.
- Material Limits: Limited to high-fluidity metals.
- Volume: Requires a large production volume to be economical.

Shell Mold Casting

- Definition: A metal casting process where the mold is a thin, hardened shell of fine sand mixed with a thermosetting resin binder, which is then backed up by another material.
- Suitability: Particularly good for steel castings under 20 lbs, but can be used for almost any metal cast in sand.
- Typical Parts: Cylinder heads, gears, bushings, connecting rods, camshafts, and valve bodies.

Shell Mold Process (Sequential Operations)

- 1. Preparation: A fine-grained sand is mixed with a thermosetting resin binder. A special metal pattern is heated (to 350°F–700°F) and coated with a parting agent (silicone).
- 2. Forming the Shell: The heated pattern is placed over a dump box containing the sand-resin mixture. The box is inverted, allowing the sand to fall and form a layer (shell) around the hot pattern.
- 3. Partial Curing: The heat from the pattern partially cures (hardens) the shell.
- 4. Final Curing: The box is righted, the top is removed, and the shell is placed in an oven to fully cure (harden) the shell.
- 5. Stripping: The rigid shell is stripped (ejected) from the pattern using ejector pins.
- 6. Assembly and Pouring: Two matched shell halves are joined, clamped, and placed in a flask (often supported by metal shot or other backing material) for pouring the molten metal.

Advantages and Disadvantages

Advantages	Disadvantages
Better accuracy and finish ($\text{Very good surface finish}$).	Limited part size (though larger parts can be made).
High production rate and low labor cost.	High equipment cost.
Can form complex shapes and fine details.	Poor material strength (in some materials).
Low tooling and equipment cost (compared to some processes).	High porosity possible.
Little scrap generated (scrap can be recycled).	

Note: There is a contradiction in the provided sources regarding "Low tooling cost" vs. "High equipment cost," and the advantage/disadvantage of "part size." The notes above reflect the common points.

Permanent Mold Casting

- Definition: A casting process involving pouring molten metal by gravity into a reusable steel (or cast iron) mold.
- Contrast: Unlike sand molds (broken after each use), a permanent mold can be used for thousands of casting cycles.
- Cost: Manufacturing the metal mold is much more expensive than sand/investment molds, requiring a minimum number of castings to be profitable.
- Materials: Casts ferrous and non-ferrous metals and alloys.

3 Steps in Permanent Mold Casting

1. Mold is preheated and coated (using a spray nozzle).
2. Cores (if used) are inserted and mold is closed (using a hydraulic cylinder).
3. Molten metal is poured into the mold, where it solidifies.

Advantages and Disadvantages

Advantages	Disadvantages
Better mechanical properties.	Costly mold (high initial cost).
Homogeneous grain structure and chemical composition.	Limited to simpler shapes only.
Low shrinkage and gas porosity.	Limited to small castings.
Good surface quality and low dimension tolerances.	Holes cannot be smaller than 1.6 mm or deeper than $1.5 \times$ diameter.
Can be used to replace die-casting where short runs are involved.	Impractical for high-volume manufacturing due to high cost/long cycle times (Note: *The slide has a conflicting point about high-volume feasibility*).

Investment Casting (Lost-Wax Process)

- Definition: A precision casting process using an expendable wax pattern to create a ceramic shell mold.
- Accuracy: Used for casting intricate forms with undercuts.

Steps in Investment Casting

1. Wax patterns are produced by injection molding.
2. Multiple patterns are assembled to a central wax sprue (forming a "tree").
3. A ceramic shell is built by immersing the assembly in a liquid ceramic slurry and coating it with fine sand (repeatedly).
4. The ceramic is dried, and the wax is melted out (hence "lost-wax" process). The ceramic shell is fired to burn off all remaining wax.
5. The shell is filled with molten metal by gravity pouring, where it solidifies into a single casting (parts, gates, sprue are one solid piece).
6. The ceramic shell is broken off by vibration or water blasting.
7. The final parts are cut away from the sprue using a high-speed saw.

Advantages and Disadvantages

Advantages	Disadvantages
Many intricate forms with undercuts can be cast.	Expensive process.

Advantages	Disadvantages
A very smooth surface is obtained (no parting line).	Requires very long production-cycle times compared to other processes.
Dimensional accuracy is good.	Impractical for high-volume manufacturing due to high cost and long cycle times.
Can cast un-machinable parts to preplanned shape.	Poor material strength (in some materials).
High porosity possible (Note: *Contradictory to the benefits expected from precision casting*).	Poor surface finish and tolerance (Note: *Contradictory to the known precision of the process*).

Investment Casting (Lost Wax)

- Principle: A wax or polymer pattern is coated several times with a refractory material slurry to form a strong shell. The wax is then melted out (lost wax), and molten metal is poured into the cavity.
- Accuracy: Has excellent dimensional accuracy and tight tolerances (e.g., 0.075 mm).
- Design: Allows casting of parts with extremely complex shapes, intricate features, and undercuts as a single piece.
- Surface: Excellent surface finish (low roughness) with no parting lines (no flash).
- Sections: Short or shallow features and thin sections (down to 0.40 mm) can be cast without cold shut defects.
- Cost: Relatively expensive due to many complex steps, high cost of dies for patterns, and specialized equipment.
- Limitations: Parts are difficult to cast if they require cores, holes smaller than 1.6 mm, or deeper than $1.5 \times$ diameter.

Centrifugal Casting Process

- Definition: Molten metal is poured into a preheated, spinning die (vertical or horizontal axis).
- Mechanism: Centrifugal force distributes the molten metal in the mold at pressures up to 100 times the force of gravity.
- Result: Components have high material soundness and are used in high-reliability applications (e.g., jet engine parts).

Advantages

- Castings acquire high density, high mechanical strength, and fine-grained structure.
- Inclusions and impurities are lighter and pushed to the inner surface (easy to machine away).
- Gates and risers are not needed.
- High output and cost-effective compared to forgings.
- Formation of hollow interiors without cores (especially in horizontal casting).

Types & Disadvantages

Type	Simple Feature	Best Suited For	Disadvantages
Horizontal Centrifugal	Die rotates about the horizontal axis. Rotational speed is high to offset gravity.	Long cylindrical parts where length $\gg$ outside diameter (e.g., straight tube sections).	Inaccurate diameter of the inner surface (due to metal distribution). Not all alloys can be cast.
Vertical	Die rotates about the	Components with O.D.	Wall thickness is a function of

Type	Simple Feature	Best Suited For	Disadvantages
Centrifugal	vertical axis.	shaping (outside diameter) for reduced post-processing.	the amount of metal poured.

Continuous Casting

- Definition: A process that allows metals to be shaped and solidified without interruption, producing a continuous length of metal (slabs, blooms, billets).
- Action: Metal is poured into a mold and rapidly chilled to ensure a uniform grain structure.
- Process Flow: Molten metal $\rightarrow$ Mold $\rightarrow$ Cooling water $\rightarrow$ Partially solidified metal is stretched $\rightarrow$ Fully solidified metal is straightened $\rightarrow$ Cut to length by saws.

Advantages

- Produces long tubes/forms in a variety of shapes and sizes.
- Castings are consistently homogeneous and dense (good for pressure applications).
- Castings are straight and concentric (no deviation).
- Less material is wasted (less machining stock needed).
- Cost-efficient for standard shapes and sizes.

Disadvantages

- High Cost of Setup: High cost of creating the mold and time spent setting up the machine.
- Not economical for small quantities or special metal alloys in smaller quantities.
- Limited to simple shapes that have a stable cross-section (not good for tedious, difficult casting projects).
